

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
from sklearn.tree import DecisionTreeClassifier , plot_tree , export_text
```

```
In [3]: df = pd.read_excel("Rework_cumulative_brand_data_cluster.xlsx")
df
```

Out[3]:

	Brand	Quantity (Grams)	MRP	Price	Price/100 Grams	Discount (%)	Cluster	projected_rating_cou
0	Happilo	500.0	265	79	15.800000	70	2	2125
1	Happilo	500.0	225	110	22.000000	51	2	436
2	Happilo	250.0	125	59	23.600000	52	2	209
3	Happilo	250.0	125	59	23.600000	52	2	209
4	Happilo	250.0	125	61	24.400000	51	2	300
...	...	...	...	...	...	...	...	...
443	The Whole Truth	210.0	1199	1009	480.476190	16	0	14
444	The Whole Truth	210.0	1199	1009	480.476190	16	0	14
445	Yogabar	750.0	4999	3818	509.066667	23	1	75
446	Yogabar	750.0	4999	3818	509.066667	23	1	75
447	Happilo	0.5	275	172	34400.000000	37	2	22

448 rows × 8 columns

```
In [5]: df.columns = df.columns.str.strip()
df.columns
```

```
Out[5]: Index(['Brand', 'Quantity ( Grams)', 'MRP', 'Price', 'Price/100 Grams',
              'Discount (%)', 'Cluster', 'projected_rating_count'],
              dtype='object')
```

```
In [6]: features = ['Quantity ( Grams)', 'Price']
target = 'Cluster'
```

```
In [9]: X = df[features]
y = df[target]
X = X.fillna(X.median())
y = y.fillna(y.mode()[0])
```

```
In [10]: clf = DecisionTreeClassifier(max_depth=4,random_state=42)
         clf.fit(X,y)
```

```
Out[10]: ▾ DecisionTreeClassifier ⓘ ?
         ▶ Parameters
```

```
In [12]: tree_rules = export_text(clf,feature_names=features)
         print(tree_rules)
```

```
|--- Price <= 488.50
|   |--- Quantity ( Grams) <= 550.00
|   |   |--- Quantity ( Grams) <= 402.50
|   |   |   |--- class: 2
|   |   |--- Quantity ( Grams) > 402.50
|   |   |   |--- Price <= 350.50
|   |   |   |   |--- class: 2
|   |   |   |--- Price > 350.50
|   |   |   |   |--- class: 0
|   |--- Quantity ( Grams) > 550.00
|   |   |--- class: 0
|--- Price > 488.50
|   |--- Price <= 1798.00
|   |   |--- Quantity ( Grams) <= 205.50
|   |   |   |--- class: 2
|   |   |--- Quantity ( Grams) > 205.50
|   |   |   |--- Price <= 531.50
|   |   |   |   |--- class: 0
|   |   |   |--- Price > 531.50
|   |   |   |   |--- class: 0
|   |--- Price > 1798.00
|   |   |--- Quantity ( Grams) <= 1502.50
|   |   |   |--- class: 1
|   |   |--- Quantity ( Grams) > 1502.50
|   |   |   |--- class: 3
```

```
In [13]: plot_tree(clf,
                   feature_names=features,
                   class_names=[f"Cluster {c}" for c in sorted(y.unique())], # Name the
                   filled=True,      # Color nodes by majority class
                   rounded=True,     # Rounded boxes
                   impurity=False,   # Hide Gini impurity to keep it cleaner to read
                   fontsize=10)

plt.title("Decision Tree Rules: Determining Clusters based on Price & Quantity", fo
plt.tight_layout()
```

Decision Tree Rules: Determining Clusters based on Price & Quantity

